

Lab Programs

13-17

22/7/26

13. Dataset: Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

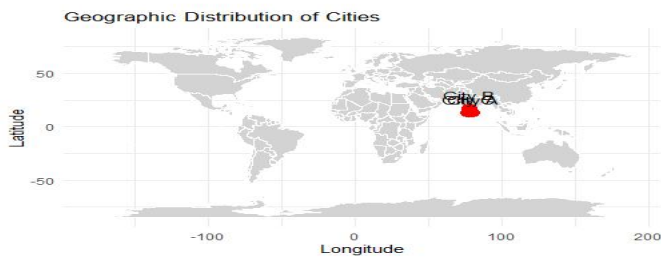
13.A:Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

Code:

```
library(ggplot2)
library(maps)
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200),
  Longitude = c(80.27, 78.48, 77.59),
  Latitude = c(13.08, 17.38, 12.97)
)
world <- map_data("world")
ggplot() +
  geom_polygon(data = world,
    aes(x = long, y = lat, group = group),
    fill = "lightgray",
    color = "white") +
  geom_point(data = geo_data,
    aes(x = Longitude, y = Latitude),
    color = "red",
    size = 4) +
  geom_text(data = geo_data,
    aes(x = Longitude, y = Latitude, label = City),
    vjust = -1,
    size = 4) +
  labs(
    title = "Geographic Distribution of Cities",
    x = "Longitude",
    y = "Latitude"
```

```
) +  
coord_fixed(1.3) +  
theme_minimal()
```

Output:

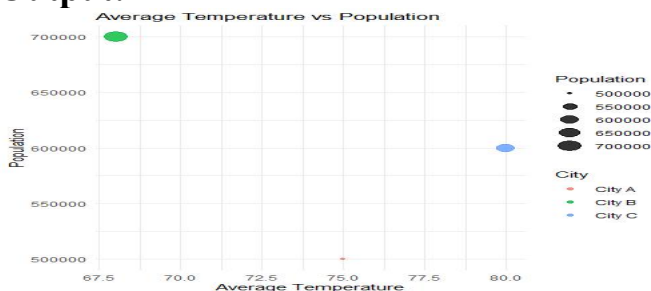


13.B:Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

Code:

```
library(ggplot2)  
geo_data <- data.frame(  
  City = c("City A","City B","City C"),  
  Population = c(500000,700000,600000),  
  Avg_Temperature = c(75,68,80)  
)  
ggplot(geo_data,  
  aes(x = Avg_Temperature,  
      y = Population,  
      color = City,  
      size = Population)) +  
  geom_point(alpha = 0.8) +  
  labs(  
    title = "Average Temperature vs Population",  
    x = "Average Temperature",  
    y = "Population"  
  ) +  
  theme_minimal()
```

Output:



13.C:Build a table to display the geographic data for each city. Label the table elements.

Code:

```
geo_data <- data.frame(  
  City = c("City A","City B","City C"),  
  Population = c(500000,700000,600000),  
  Avg_Temperature = c(75,68,80),  
  Elevation = c(1000,800,1200)  
)  
print(geo_data)
```

Output:

	City	Population	Avg_Temperature	Elevation
1	City A	5e+05	75	1000
2	City B	7e+05	68	800
3	City C	6e+05	80	1200

13.D:..Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Code:

```
library(shiny)  
library(ggplot2)  
geo_data <- data.frame(  
  City = c("City A","City B","City C"),  
  Population = c(500000,700000,600000),  
  Avg_Temperature = c(75,68,80),  
  Elevation = c(1000,800,1200)  
)  
ui <- fluidPage(  
  titlePanel("Geographic Data Dashboard"),  
  fluidRow(  
    column(6,  
      plotOutput("mapPlot")  
    ),  
    column(6,  
      plotOutput("scatterPlot")  
    )  
  ),  
  br(),  
  tableOutput("dataTable"))  
server <- function(input, output){  
  output$mapPlot <- renderPlot({
```

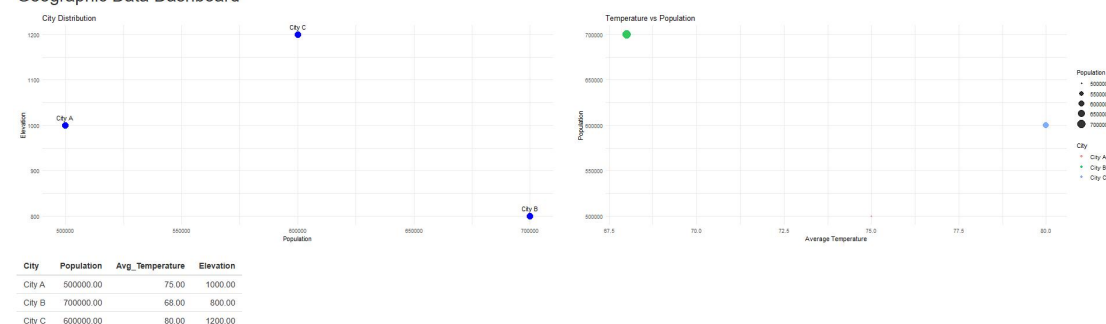
```

ggplot(geo_data,
      aes(x = Population,
          y = Elevation,
          label = City)) +
  geom_point(size = 5, color = "blue") +
  geom_text(vjust = -1) +
  labs(
    title = "City Distribution",
    x = "Population",
    y = "Elevation"
  ) +
  theme_minimal()
})
output$scatterPlot <- renderPlot({
  ggplot(geo_data,
        aes(x = Avg_Temperature,
            y = Population,
            color = City,
            size = Population)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Temperature vs Population",
    x = "Average Temperature",
    y = "Population"
  ) +
  theme_minimal()
})
output$dataTable <- renderTable({
  geo_data
})
}
shinyApp(ui, server)

```

Output:

Geographic Data Dashboard



14.A: Dataset: Survey Results

Respondent Question 1 Question 2 Question 3

1 A B C

2 B A D

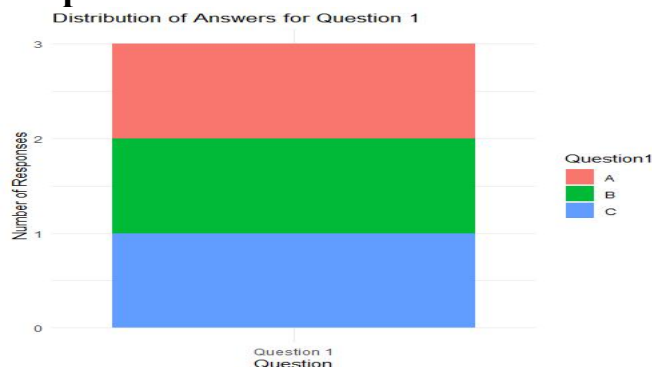
3 C A B

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

Code:

```
library(ggplot2)
survey_data <- data.frame(
  Respondent = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
ggplot(survey_data,
  aes(x = "Question 1",
    fill = Question1)) +
  geom_bar() +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Question",
    y = "Number of Responses"
  ) +
  theme_minimal()
```

Output:



14.B: Generate a radar chart to visualize the overall pattern of responses across all three questions.

Code:

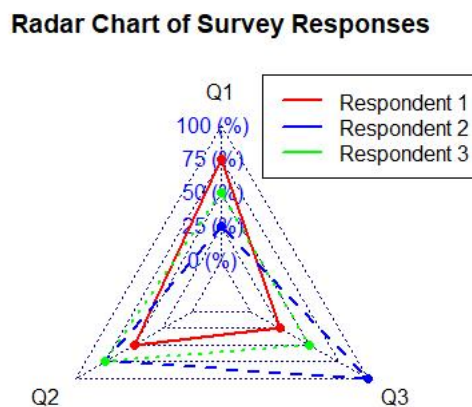
```
library(fmsb)
survey_scores <- data.frame(
  Q1 = c(3,1,2),
  Q2 = c(2,3,3),
```

```

Q3 = c(1,4,2)
)
radar_data <- rbind(
  c(4,4,4),
  c(0,0,0),
  survey_scores
)
radarchart(
  radar_data,
  axistype = 1,
  pcol = c("red","blue","green"),
  plwd = 2,
  title = "Radar Chart of Survey Responses"
)
legend("topright",
  legend = c("Respondent 1","Respondent 2","Respondent 3"),
  col = c("red","blue","green"),
  lty = 1,
  lwd = 2)

```

Output:



14.C: Build a table to show the survey response data for each respondent. Label the table elements.

Code:

```

survey_data <- data.frame(
  Respondent = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
print(survey_data)

```

Output:

Respondent					Question1	Question2	Question3
1	1	A	B	C			
2	2	B	A	D			
3	3	C	A	B			

14.D: Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

Code:

```
library(shiny)
library(ggplot2)
survey_data <- data.frame(
  Respondent = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
ui <- fluidPage(
  titlePanel("Survey Results Dashboard"),
  fluidRow(
    column(6,
      plotOutput("barPlot")),
    column(6,
      tableOutput("surveyTable"))
  )
)
server <- function(input, output){
  output$barPlot <- renderPlot({
    ggplot(survey_data,
      aes(x = "Question 1",
        fill = Question1)) +
    geom_bar() +
    labs(
      title = "Distribution of Question 1 Responses",
      x = "Question",
      y = "Count"
    ) +
    theme_minimal()
  })
  output$surveyTable <- renderTable({
    survey_data
  })
}
```

```
shinyApp(ui, server)
```

Output:

Survey Results Dashboard



Respondent	Question1	Question2	Question3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

Product ID Product Name January Sales February Sales March Sales

1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

15.A: Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Code:

```
library(ggplot2)
sales_long <- data.frame(
  Product = c("Product A","Product A","Product A",
              "Product B","Product B","Product B",
              "Product C","Product C","Product C"),
  Month = c("January","February","March",
            "January","February","March",
            "January","February","March"),
  Sales = c(2000,2200,2400,
            1500,1800,1600,
            1200,1400,1100)
)
ggplot(sales_long,
  aes(x = Product,
      y = Sales,
      fill = Month)) +
  geom_bar(stat = "identity",
    position = "dodge") +
  labs(
    title = "Quarterly Product Sales",
    x = "Product",
```

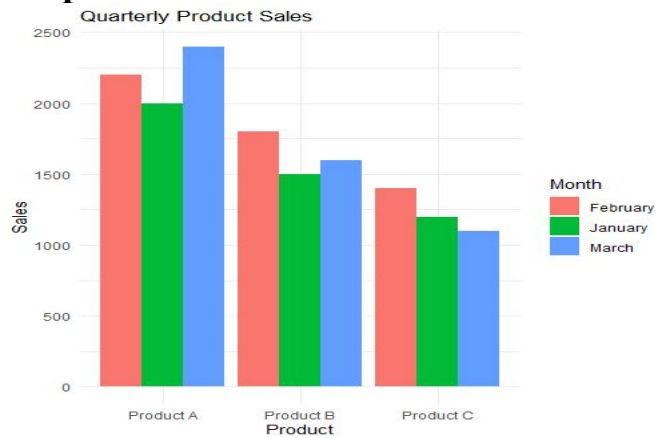


```

    y = "Sales"
  ) +
  theme_minimal()

```

Output:



15.B:Generate a stacked area chart to represent the overall sales trend for all products over the three months.

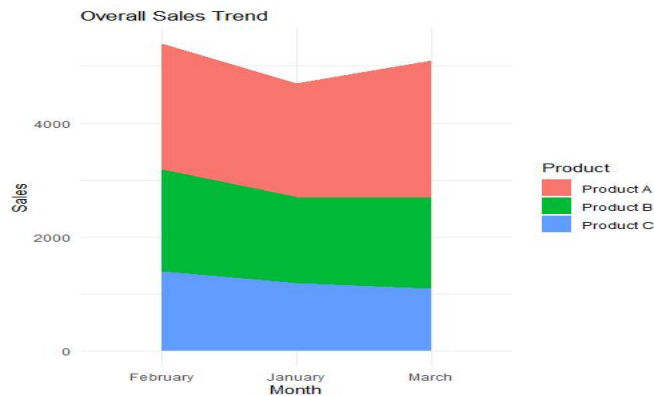
Code:

```

library(ggplot2)
library(reshape2)
sales_data <- data.frame(
  Month = c("January", "February", "March"),
  ProductA = c(2000, 2200, 2400),
  ProductB = c(1500, 1800, 1600),
  ProductC = c(1200, 1400, 1100)
)
sales_long <- melt(sales_data,
  id.vars = "Month",
  variable.name = "Product",
  value.name = "Sales")
ggplot(sales_long,
  aes(x = Month,
    y = Sales,
    fill = Product,
    group = Product)) +
  geom_area() +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()

```

Output:



15.C:Build a table to show the monthly sales data for each product. Label the table elements.

Code:

```
sales_data <- data.frame(
  Product_ID = c(1,2,3),
  Product_Name = c("Product A","Product B","Product C"),
  January_Sales = c(2000,1500,1200),
  February_Sales = c(2200,1800,1400),
  March_Sales = c(2400,1600,1100)
)
print(sales_data)
```

Output:

	Product_ID	Product_Name	January_Sales	February_Sales	March_Sales
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

15.D:Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Code:

```
library(shiny)
library(ggplot2)
sales_long <- data.frame(
  Product = c("Product A","Product A","Product A",
             "Product B","Product B","Product B",
             "Product C","Product C","Product C"),
  Month = c("January","February","March",
            "January","February","March",
            "January","February","March"),
  Sales = c(2000,2200,2400,
            1500,1800,1600,
```

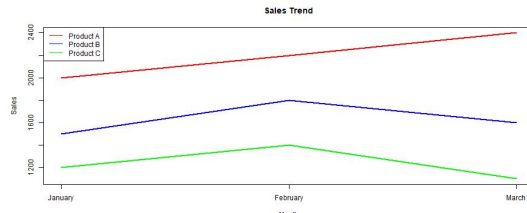
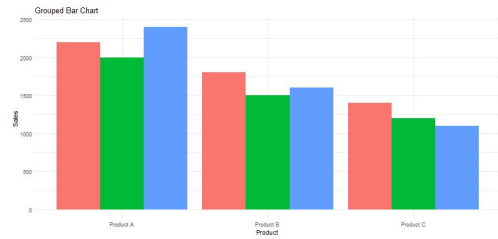
```

      1200,1400,1100)
    )
    sales_table <- data.frame(
      Product_ID = c(1,2,3),
      Product_Name = c("Product A","Product B","Product C"),
      January = c(2000,1500,1200),
      February = c(2200,1800,1400),
      March = c(2400,1600,1100)
    )
    area_data <- data.frame(
      Month = factor(c("January","February","March"),
        levels = c("January","February","March")),
      ProductA = c(2000,2200,2400),
      ProductB = c(1500,1800,1600),
      ProductC = c(1200,1400,1100)
    )
    ui <- fluidPage(
      titlePanel("Product Sales Dashboard"),
      fluidRow(
        column(6, plotOutput("bar")),
        column(6, plotOutput("area"))
      ),
      hr(),
      tableOutput("table")
    )
    server <- function(input, output){
      output$bar <- renderPlot({
        ggplot(sales_long,
          aes(Product, Sales, fill = Month)) +
          geom_bar(stat = "identity", position = "dodge") +
          labs(title = "Grouped Bar Chart") +
          theme_minimal()
      })
    }
  }
}

```

Output:

Product Sales Dashboard



Product_ID	Product_Name	January	February	March
1.00	Product A	2000.00	2200.00	2400.00
2.00	Product B	1500.00	1800.00	1600.00
3.00	Product C	1200.00	1400.00	1100.00

Set 16: Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID Age Gender Income (in \$)

1 28 Female 50000

2 35 Male 60000

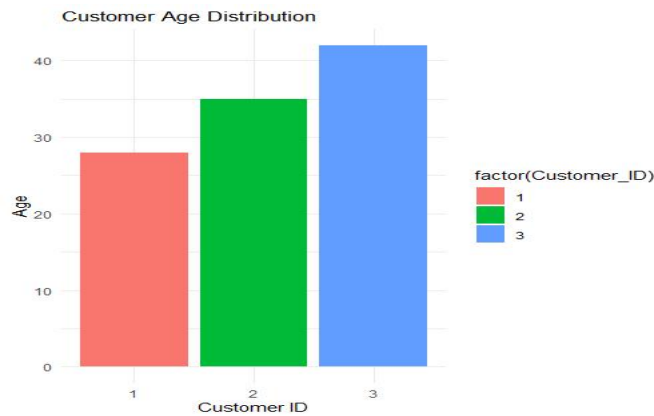
3 42 Female 75000

16.A: Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

Code:

```
library(ggplot2)
customer_data <- data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
ggplot(customer_data,
  aes(x = factor(Customer_ID),
    y = Age,
    fill = factor(Customer_ID))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age"
  ) +
  theme_minimal()
```

Output:

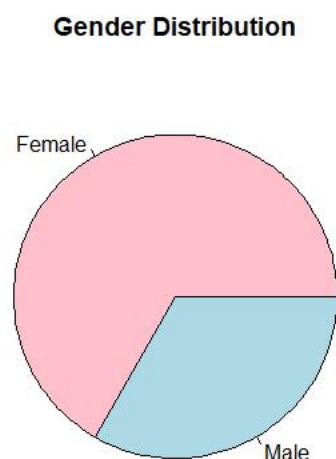


16.B:Generate a pie chart to represent the distribution of customers by gender.

Code:

```
customer_data <- data.frame(
  Gender = c("Female","Male","Female")
)
gender_count <- table(customer_data$Gender)
pie(
  gender_count,
  labels = names(gender_count),
  col = c("pink","lightblue"),
  main = "Gender Distribution"
)
```

Output:



16.C:Build a table to show the demographic information for each customer. Label the table elements.

Code:

```
customer_data <- data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female")
)
```

```

Gender = c("Female","Male","Female"),
Income = c(50000,60000,75000)
)
print(customer_data)

```

Output:

Customer_ID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

16.D: Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Code:

```

library(shiny)
library(ggplot2)
customer_data <- data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
ui <- fluidPage(
  titlePanel("Customer Demographics Dashboard"),
  fluidRow(
    column(6,
      plotOutput("barPlot")),
    column(6,
      plotOutput("piePlot"))
  ),
  hr(),
  tableOutput("customerTable")
)
server <- function(input, output){
  output$barPlot <- renderPlot({
    ggplot(customer_data,
      aes(x = factor(Customer_ID),
          y = Age,
          fill = factor(Customer_ID))) +
    geom_bar(stat = "identity") +
    labs(

```

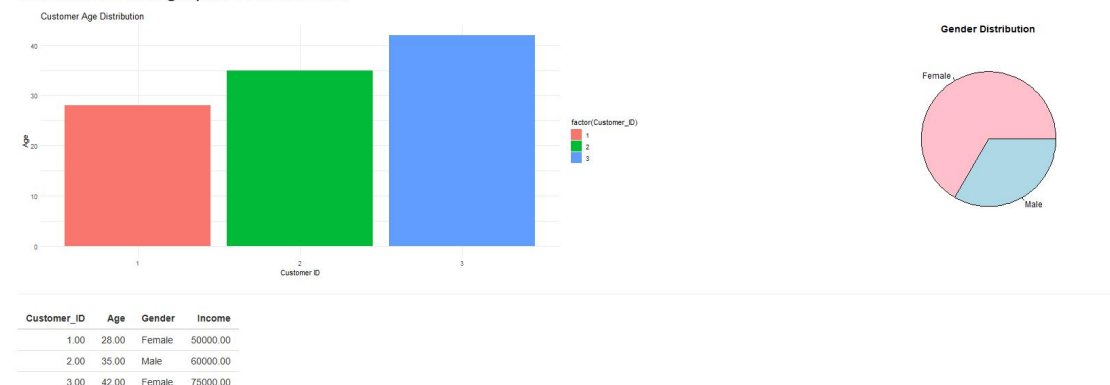
```

    title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age"
  ) +
  theme_minimal()
})
output$piePlot <- renderPlot({
  gender_count <- table(customer_data$Gender)
  pie(
    gender_count,
    labels = names(gender_count),
    col = c("pink", "lightblue"),
    main = "Gender Distribution"
  )
})
output$customerTable <- renderTable({
  customer_data
})
}
shinyApp(ui, server)

```

Output:

Customer Demographics Dashboard



Set 17: Employee Performance Analysis

Dataset: Employee Performance

Employee ID Department Years of Service Performance Score

1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

17.A: Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Code:

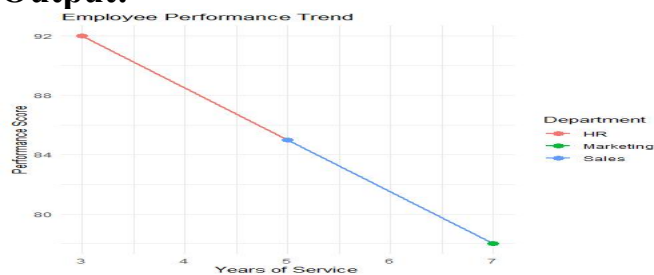
```
library(ggplot2)
```

```

employee_data <- data.frame(
  Employee_ID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  Years_of_Service = c(5,3,7),
  Performance_Score = c(85,92,78)
)
ggplot(employee_data,
  aes(x = Years_of_Service,
      y = Performance_Score,
      color = Department,
      group = 1)) +
  geom_line(size = 1) +
  geom_point(size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()

```

Output:



17.B:Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Code:

```

library(ggplot2)
employee_data <- data.frame(
  Department = c("Sales","HR","Marketing")
)
ggplot(employee_data,
  aes(x = Department,
      fill = Department)) +
  geom_bar() +
  labs(
    title = "Employees by Department",
    x = "Department",
    y = "Number of Employees"
  ) +

```



```
theme_minimal()
```

Output:



17.C: Build a table to display the performance data for each employee. Label the table elements.

Code:

```
employee_data <- data.frame(  
  Employee_ID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  Years_of_Service = c(5,3,7),  
  Performance_Score = c(85,92,78)  
)
```

```
print(employee_data)
```

Output:

Employee	ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

17.D: Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Code:

```
library(shiny)  
library(ggplot2)  
employee_data <- data.frame(  
  Employee_ID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  Years_of_Service = c(5,3,7),  
  Performance_Score = c(85,92,78)  
)  
ui <- fluidPage(  

```

```

titlePanel("Employee Performance Dashboard"),
fluidRow(
  column(6,
    plotOutput("linePlot")),
  column(6,
    plotOutput("barPlot"))),
hr(),
tableOutput("employeeTable")
)
server <- function(input, output){
  output$linePlot <- renderPlot({
    ggplot(employee_data,
      aes(x = Years_of_Service,
          y = Performance_Score,
          color = Department,
          group = 1)) +
    geom_line(size = 1) +
    geom_point(size = 3) +
    labs(
      title = "Employee Performance Trend",
      x = "Years of Service",
      y = "Performance Score"
    ) +
    theme_minimal()
  })
  output$barPlot <- renderPlot({
    ggplot(employee_data,
      aes(x = Department,
          fill = Department)) +
    geom_bar() +
    labs(
      title = "Employees by Department",
      x = "Department",
      y = "Number of Employees"
    ) +
    theme_minimal()
  })
  output$employeeTable <- renderTable({
    employee_data
  })
}

shinyApp(ui, server)

```

Output:

Employee Performance Dashboard

